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MATE: Asynchronous Multi-Rate Diffusion Policy Composition

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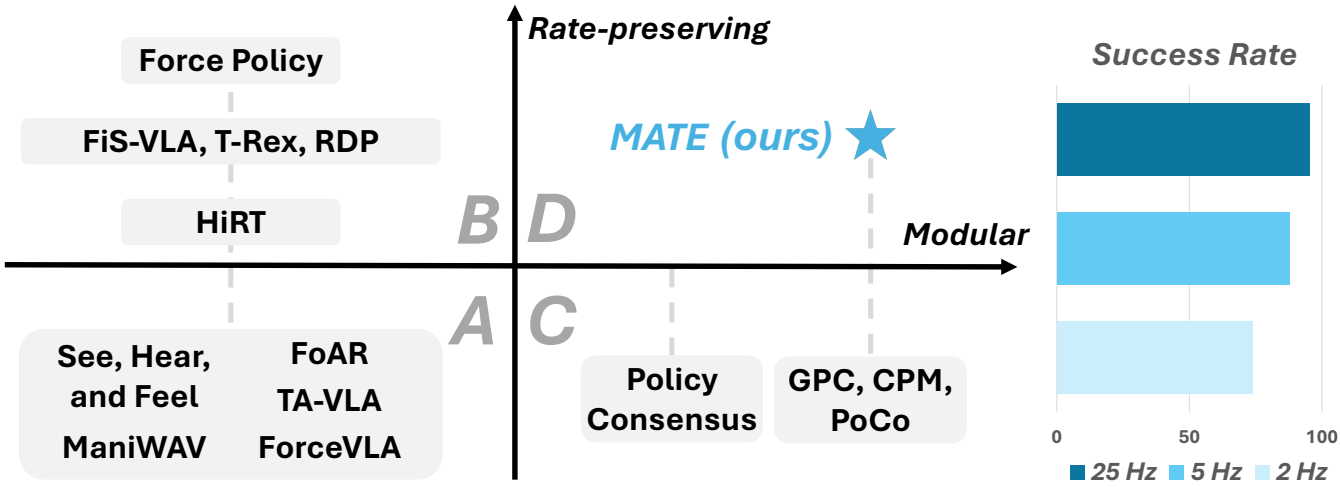


Fig. 1: MATE enables modular, rate-preserving policy composition. ***(Left)*** We organize representative manipulation methods along two complementary properties: *modularity* and *rate preservation*. (A) Unified policies are neither modular nor rate-preserving. (B) Slow-fast and hierarchical methods preserve heterogeneous rates but rely on coupled architectures. (C) Modular composition methods combine independently trained components but synchronize them at a shared inference rate. (D) MATE achieves both properties through asynchronous multi-rate policy composition. ***(Right)*** As a motivating experiment, we evaluate a standalone force policy on the *Flip* task across five different objects, with the gripper initialized near the object in a pre-contact configuration. Higher policy inference rates consistently improve task success, motivating the need to preserve heterogeneous inference rates during policy composition.

***Abstract*— Diffusion policies have shown strong potential for robotic imitation learning, but different policy components can benefit from substantially different inference rates. For example, visual policies provide rich global guidance but are computationally expensive, whereas policies conditioned on low-dimensional signals can be queried more frequently for reactive correction. However, existing diffusion policies either synchronize all components to the slowest one or rely on jointly trained, task-specific multi-rate architectures that are difficult to extend. We propose MATE, a modular framework for asynchronous multi-rate diffusion policy composition. MATE allows independently trained component policies to operate at their native inference rates and contribute denoising guidance whenever new predictions become available. To consistently compose predictions generated at different robot states, we derive a reference-frame rebasing rule for diffusion variables under relative action representations and align delayed guidance before fusion. We instantiate MATE in contact-rich manipulation by composing a low-frequency vision policy with a high-frequency force policy, and further demonstrate its generality through asynchronous composition of complementary 3D and 2D visual policies. Across multiple real-world tasks, MATE outperforms various baselines, demonstrating benefits beyond high-frequency inference alone.**

I. Introduction

Diffusion policies [8, 37] have emerged as a powerful paradigm for robotic imitation learning, generating expressive action trajectories through iterative denoising [19, 36]. Recent studies extend diffusion policies with additional sensory and task information, including point clouds, language instructions, force/torque measurements, tactile signals, and audio, to improve manipulation across diverse settings [17, 21, 29, 40, 41]. However, existing work primarily focuses on *what information* should be incorporated, while paying less attention to *how frequently* different policy components should be queried during closed-loop execution.

These considerations motivate two desirable properties for combining policy components:

- **Modular.** Modularity enables independently trained policy components to be reused and composed without joint retraining or redesigning the architecture. This supports incremental extension: new sensing capabilities or specialized behaviors can be incorporated by adding a complementary policy while retaining existing components. For example, an existing VLA policy could be augmented with a specialized force-conditioned policy for contact-aware correction *without extra retraining*.
- **Rate-preserving.** Different policy components may naturally operate at different inference rates due to their sensing characteristics. Our motivated experiment in Fig. 1 shows that higher policy inference frequency improves contact-stage success. This highlights the importance of preserving the native inference rate of each policy, allowing fast components to provide frequent corrections without being bottlenecked by slow ones.

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Existing approaches do not fully satisfy both properties. As summarized in Fig. 1, unified policies [17, 23, 29, 43, 45] couple multiple inputs within a shared inference pipeline, and do not explicitly preserve heterogeneous inference rates. Slow-fast and hierarchical architectures [5, 13, 34, 41, 44] preserve heterogeneous update rates by introducing fast reactive components alongside slower ones. However, these components are typically designed as coupled architectures, making it difficult to incorporate additional independently trained policies without architectural modification.

Policy composition offers a more modular alternative by combining independently trained policies at inference time [4, 6, 27, 38]. However, existing composition methods generally perform synchronous composition, where participating components are evaluated and combined at common inference steps. They therefore do not explicitly preserve heterogeneous component rates, and the composed system is effectively constrained by the slower components. As illustrated by the motivating experiment in Fig. 1, this can be undesirable when a lightweight policy benefits from more frequent inference. This synchronization assumption breaks in a multi-rate setting: a slow policy prediction may be reused while a fast policy is queried multiple times, causing their action trajectories and denoising guidance to correspond to different timestamps and potentially different action reference frames. Directly composing such predictions is geometrically inconsistent, while waiting for all policies would again reduce execution to the slowest inference rate.

To address this problem, we propose MATE, a framework for asynchronous multi-rate diffusion policy composition. MATE decouples component-policy inference from action composition, allowing each policy to operate at its native rate and contribute denoising guidance whenever a new prediction becomes available. To consistently compose guidance generated at different robot states, we derive a principled reference-frame rebasing rule for clean actions, noisy diffusion variables, and predicted noise under relative action representations. The aligned guidance is then composed asynchronously, enabling high-frequency action updates without repeatedly querying slow policy components or requiring joint retraining.

We instantiate MATE in contact-rich manipulation by composing a low-frequency vision policy with a high-frequency force/torque policy. The vision policy supplies global spatial and long-horizon task guidance, while the force policy provides rapid local corrections during physical interaction. We further demonstrate that the framework is not restricted to vision-force fusion by composing complementary 3D and 2D visual policies in an occluded pick-and-place task. Across the evaluated tasks, MATE outperforms synchronous composition, specially designed multimodal policies, and a matched-frequency end-to-end baseline. Our main contributions are summarized as follows:

- We formulate *asynchronous multi-rate diffusion policy composition*, allowing independently trained policy components to operate at its native inference rates without being bottlenecked by the slowest component.
- We derive a principled reference-frame rebasing rule for diffusion variables and combine it with temporal overlap selection, enabling composition of denoising guidance generated at different query times and robot states.
- We demonstrate the framework through multi-rate vision-force and 3D-2D vision policy composition, and show improved performance over baselines.

II. Related Works

A. Policy Composition in Robotics

Policy composition has been widely explored as a way to combine multiple generative models, constraints, or guidance signals at inference time. Related ideas have appeared in visual generation [7, 10, 15, 16, 26], language modeling [22, 24, 31, 42], and planning [11, 32, 33], where different semantic or structural conditions are composed to guide generation. In robotics, recent works extend this idea to policy learning. PoCo [38] composes diffusion policies trained from heterogeneous robot datasets, while General Policy Composition [4] combines diffusion- or flow-based robot policies through test-time distribution-level composition. In addition, Policy Consensus [6] learns router weights to adaptively combine modality-specific guidance at inference time.

However, existing robot policy composition methods are frequency-agnostic [4, 6, 25, 38]. They assume that all modalities are queried at the same time, in the same reference frame. This assumption simplifies composition, but it also implicitly requires all policies to operate at a shared decision frequency. As a result, the composed policy is often limited by the slowest modality or policy component. Contact-rich manipulation is a representative example where this synchronization assumption becomes problematic. Stable physical interaction always requires fast local corrections from force/torque feedback. MATE addresses this gap through asynchronous multi-rate composition, allowing independently trained policies to contribute guidance at their native inference rates while aligning their predictions in time and action reference frame.

B. Policy Inference Frequency and Multi-Rate Execution

Policy inference frequency determines how often new observations can influence action predictions, and is distinct from sensor sampling and low-level control rates. Prior work improves policy responsiveness through consistency distillation [14, 35] or by reusing previous predictions to reduce denoising computation [12, 20]. Other approaches address execution latency through observation-action latency matching [9] or asynchronous action chunk generation with continuity constraints [1]. These methods improve inference efficiency and execution timing for individual policies.

Multi-rate architectures provide another way to improve responsiveness by operating different components at different temporal scales. This is particularly useful in contact-rich manipulation, where rapidly changing interaction forces can require frequent feedback-driven corrections [13, 17, 21, 28, 41]. For example, Reactive Diffusion Policy (RDP) combines a slow latent diffusion policy with a faster tactile-conditioned

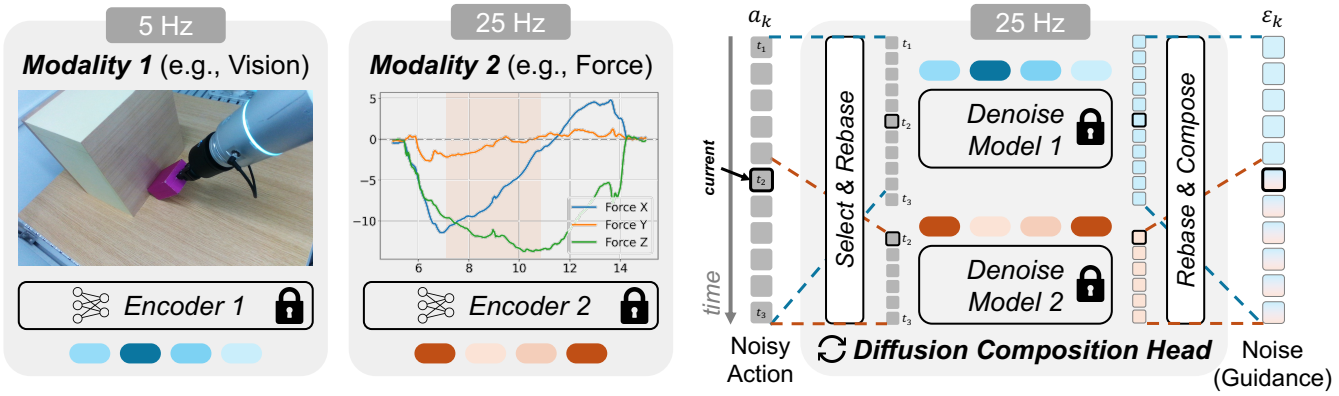


Fig. 2: Overview of the MATE Framework. MATE asynchronously composes component diffusion policies operating at heterogeneous inference rates. ***(Left)*** We illustrate vision-force composition as an example, where a lower-rate vision-conditioned policy provides long-horizon scene guidance and a higher-rate force-conditioned policy provides local contact guidance. ***(Right)*** In the diffusion composition head, the overlapping noisy action segment is rebased into the current force frame for force denoising. The corresponding vision guidance is rebased to the same frame, fused with force guidance, and mapped back to update the full-horizon noise prediction.

decoder [41], while Force Policy uses a slower global vision policy together with a higher-rate local force policy [13]. Related slow-fast and hierarchical architectures likewise employ lightweight reactive components between slower high-level updates [5, 34, 44]. These works demonstrate the benefit of heterogeneous inference rates, but their components are typically designed and trained as coupled architectures rather than independently reusable policies. MATE combines this rate-preserving capability with the modular policy composition setting discussed above, allowing independently trained policies to contribute asynchronously at their native inference rates without being bottlenecked by the slowest component.

III. Method

We propose MATE, a general framework for asynchronous multi-rate diffusion policy composition. The framework composes independently trained component policies that may use different sensing modalities, observation representations, and inference rates, without synchronizing their execution to the slowest component. Each policy is queried according to its own inference schedule, allowing newly available guidance from a faster policy to update the composed action while reusing guidance from slower policies. Fig. 2 illustrates a vision-force instantiation: a lower-rate vision policy provides global, long-horizon guidance, while a higher-rate force/torque policy supplies local contact feedback.

At inference time, MATE maintains the latest denoising guidance from each component policy and composes their temporally overlapping predictions during diffusion sampling. Unlike synchronous composition (§III-A), MATE aligns predictions from different query times by rebasing diffusion variables into a common action reference frame (§III-B), then composes the aligned guidance over temporally overlapping horizons (§III-C).

A. Preliminary: Synchronous Diffusion Policy Composition

We first review the standard setting of diffusion policy composition [4, 38]. Let $a_0 \in \mathbb{R}^{H\times d}$ denote a clean action trajectory over horizon H and dimension d. A diffusion policy defines a noisy action as

$$a_k = \sqrt{\bar{\alpha}_k} a_0 + \sqrt{1-\bar{\alpha}_k}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{1}$$

where k is the diffusion timestep and $\bar{\alpha}_k$ is the cumulative noise schedule. Consider a collection of M component diffusion policies,$\pi^{(m)}{}_{m=1}^{M}$, that share a compatible action space and diffusion parameterization.Each policy may use its own observation representation and conditioning pipeline. Given the shared noisy trajectory a_k, policy m predicts

$$\epsilon_k^{(m)} = \pi^{(m)}\left(a_k, k; c^{(m)}\right) \qquad m \in 1, \ldots, M, \tag{2}$$

where $c^{(m)}$ denotes the policy-specific condition after observation encoding. Component policies may use different sensing modalities, distinct representations of the same modality, or complementary subsets of the available observations.

In the synchronous setting, all component policies are queried at the same physical time, and their predictions are defined over the same action horizon and in the same action reference frame. Their noise predictions can therefore be directly composed as

$$\epsilon_k = \sum_{m=1}^{M} \frac{w^{(m)}}{\sum_{j=1}^{M} w^{(j)}} \epsilon_k^{(m)}, \tag{3}$$

where $w^{(m)} \geq 0$ is the composition weight of the m-th policy and $\sum_{m=1}^{M} w^{(m)} > 0$. The composed noise prediction is then used in a standard DDIM [36] update $a_{k-1} = \mathrm{DDIMStep}\left(a_k, \epsilon_k, k\right)$. This synchronous composition serves as the basis for our asynchronous extension.

B. Reference-Frame Rebase for Diffusion Variables

In asynchronous composition, the component policies may be queried at different physical times. Consequently, their predicted action trajectories and denoising guidance may be expressed in different action reference frames. Before composition, all available diffusion variables must therefore be transformed into a common reference frame.

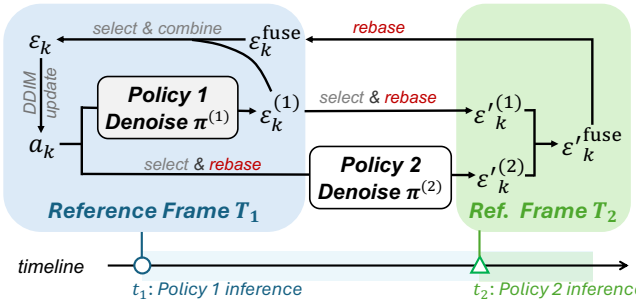


Fig. 3: Pairwise Asynchronous Policy Composition with Reference-Frame Rebasing. We illustrate the composition of two component policies queried at different times. Policy 1 provides full-horizon guidance in frame T_1, while Policy 2 is queried later and produces guidance in frame T_2. MATE selects the temporally overlapping noisy action segment and rebases it from T_1 to T_2 for Policy 2 denoising. The corresponding Policy 1 guidance is also selected and rebased to T_2, where the aligned guidance from both policies is fused. The fused guidance is then mapped back to T_1 and inserted into the full-horizon diffusion prediction.

We use relative action chunks [8, 9, 39, 46], where each predicted action is defined with respect to the robot state at the corresponding policy query time. Relative actions provide a convenient representation for closed-loop control, but they are inherently tied to their initial reference state: the same relative action can induce different physical motions when executed from different robot poses. Therefore, directly composing predictions generated at different query times would mix diffusion variables expressed in different action frames.

Consider an arbitrary pair of policy query events at times t_1 and t_2, with $t_1 \leq t_2$. A cached prediction generated at t_1 is expressed in the action frame associated with the robot pose at t_1, whereas newly available guidance at t_2 is expressed in the frame associated with the robot pose at t_2. We derive a pairwise rebase operation that transforms diffusion variables from the source frame at t_1 to the target frame at t_2. This operation is independent of the observation modality or policy architecture and can be applied separately to every cached policy prediction before composition.

Let T_1 and T_2 denote the robot poses at t_1 and t_2, respectively. Then, the transformation from the old action frame at t_1 to the current action frame at t_2 is $T_{2\leftarrow 1} = T_2^{-1}T_1$. Let $T_{2\leftarrow 1} = (R_{2\leftarrow 1}, p_{2\leftarrow 1})$, where $R_{2\leftarrow 1}$ denotes the rotation matrix and $p_{2\leftarrow 1}$ is the translational offset.

Action Representation and Transformation. Since composition is performed in the noise space, the rebase operation must be compatible with clean actions, noisy samples, and predicted noises. We thus need an action representation whose frame transformation has an orthogonal linear part and a separate affine offset. The linear part can be applied to noises without changing their Gaussian covariance, while the offset must be handled separately for clean actions and noisy samples.

We first consider the rotation component. Let the rotation component of a relative action with respect to frame T_1 be $R = [r_1, r_2, r_3] \in SO(3)$, where $r_i \in \mathbb{R}^3$ is the i-th column. Under the transformation $T_{2\leftarrow 1}$, the same relative rotation is expressed in frame T_2 as $R' = R_{2\leftarrow 1}R$, so we have $r'_i = R_{2\leftarrow 1}r_i$. This column-wise transformation motivates the column-based 6D representation [47], $R_{6\mathrm{d}} = [r_1^\top, r_2^\top]^\top$, whose rebase is linear: $R'_{6\mathrm{d}} = \mathrm{blockdiag}\,(R_{2\leftarrow 1}, R_{2\leftarrow 1}) \cdot R_{6\mathrm{d}}$. Other action dimensions can also be handled in the same representation space. For the relative translation $x \in \mathbb{R}^3$, the clean-action rebase is affine: $x' = R_{2\leftarrow 1}x + p_{2\leftarrow 1}$. The gripper command $g \in \mathbb{R}$ is frame-invariant, so $g' = g$. Thus, each action step is represented as $a = [x^\top, R_{6\mathrm{d}}^\top, g]^\top \in \mathbb{R}^{10}$, and the full action-space transformation is

$$a' = A_{2\leftarrow 1}a + b_{2\leftarrow 1}, \tag{4}$$

where

$$\begin{cases} A_{2\leftarrow 1} = \mathrm{blockdiag}\,(R_{2\leftarrow 1}, R_{2\leftarrow 1}, R_{2\leftarrow 1}, 1) \in \mathbb{R}^{10\times 10}, \\ b_{2\leftarrow 1} = \left[p_{2\leftarrow 1}, \mathbf{0}_3^\top, \mathbf{0}_3^\top, 0\right]^\top \in \mathbb{R}^{10}. \end{cases} \tag{5}$$

Since $R_{2\leftarrow 1} \in SO(3)$, the linear part $A_{2\leftarrow 1}$ is orthogonal. Next, we show that this affine rebase leads to different rules for different diffusion variables: clean actions use the full affine transform, noisy samples include a scaled offset, and noises are transformed only by the orthogonal linear part $A_{2\leftarrow 1}$.

Diffusion Variables Transformation. We now derive how the affine rebase applies to different diffusion variables. For the clean actions, they naturally follow the full affine transform in Eq. (5). Given a noisy sample a_k and a predicted noise ϵ_k in frame T_1, the clean action estimate is

$$\hat{a}_0 = \frac{1}{\sqrt{\bar{\alpha}_k}}a_k - \frac{\sqrt{1-\bar{\alpha}_k}}{\sqrt{\bar{\alpha}_k}}\epsilon_k. \tag{6}$$

Applying the clean action rebase in Eq. (5) gives

$$\begin{aligned} \hat{a}'_0 &= A_{2\leftarrow 1}\hat{a}_0 + b_{2\leftarrow 1} \\ &= \frac{1}{\sqrt{\bar{\alpha}_k}}\left(A_{2\leftarrow 1}a_k + \sqrt{\bar{\alpha}_k}b_{2\leftarrow 1}\right) - \frac{\sqrt{1-\bar{\alpha}_k}}{\sqrt{\bar{\alpha}_k}}A_{2\leftarrow 1}\epsilon_k. \end{aligned} \tag{7}$$

Comparing Eq. (7) with the standard estimator in frame T_2, which has the same form as Eq. (6), we can obtain the rebase rules:

$$\begin{cases} a'_k = A_{2\leftarrow 1}a_k + \sqrt{\bar{\alpha}_k}b_{2\leftarrow 1}, \\ \epsilon'_k = A_{2\leftarrow 1}\epsilon_k. \end{cases} \tag{8}$$

Thus, the affine offset appears in the noisy sample with a factor $\sqrt{\bar{\alpha}_k}$, but it is not added to the diffusion noise. The noise is transformed only by the orthogonal linear part $A_{2\leftarrow 1}$, preserving the standard Gaussian diffusion distribution.

The derivation above defines a pairwise rebase operator between two action frames and does not assume a particular policy or observation modality. More generally, suppose that composition is performed at time t_c in the current action frame T_c, and let $\tau_m \leq t_c$ denote the query time of the most recent available prediction from component policy $\pi^{(m)}$. Its prediction is associated with the action frame $T_m = T(\tau_m)$. We independently compute $T_{c\leftarrow m} = T_c^{-1}T_m$ and apply the rebase rules in Eq. (8) to transform the corresponding diffusion variables into T_c. After this operation is applied

to every available component prediction, all guidance terms are expressed in the same action reference frame and can be composed consistently.

C. Asynchronous Multi-Rate Policy Composition

MATE supports the asynchronous composition of M component diffusion policies $\{\pi^{(m)}\}_{m=1}^{M}$ operating at heterogeneous inference rates. At a composition time t_c, the latest available prediction from each policy may have been generated at a different query time, over a different temporal horizon, and in a different action reference frame. We first derive asynchronous fusion for an arbitrary pair of component policies and then extend the formulation to multi-policy composition.

Pairwise asynchronous fusion. Consider two arbitrary component policies $\pi^{(1)}$ and $\pi^{(2)}$, queried at times t_1 and t_2, respectively, where $t_1 < t_2$. Their predictions are expressed in the corresponding action frames T_1 and T_2. Fig. 3 illustrates the asynchronous composition of these two arbitrary component policies. Policy $\pi^{(1)}$ provides full-horizon guidance in frame T_1, while Policy $\pi^{(2)}$ is queried later and produces guidance over an overlapping horizon in frame T_2. The following derivation applies to any pair of component policies with compatible action spaces and overlapping prediction horizons, regardless of their observation modalities, representations, or inference rates.

Let $S(\cdot)$ denote the temporal selection operator that extracts the segment of Policy 1's prediction overlapping with the horizon of Policy 2. Before querying Policy 2, we rebase this selected noisy segment into Policy 2's action frame T_2 using Eq. (8). Policy 2 then predicts

$$\epsilon_k'^{(2)} = \pi^{(2)}\left(A_{2\leftarrow 1} S(a_k) + \sqrt{\bar{\alpha}_k} b_{2\leftarrow 1}, k; c^{(2)}\right), \quad (9)$$

where $c^{(2)}$ is Policy 2's condition at time t_2. The rebase parameters are applied to every action step.

Policy 1 predicts full-horizon guidance $\epsilon_k^{(1)} = \pi^{(1)}(a_k, k; c^{(1)})$ in frame T_1, where $c^{(1)}$ is Policy 1's condition at time t_1. To fuse it with Policy 2's guidance, we select the same temporal segment and rebase Policy 1's guidance into frame T_2:

$$\epsilon_k'^{(1)} = A_{2\leftarrow 1} S(\epsilon_k^{(1)}). \quad (10)$$

Extension to multiple component policies. The transformation above is pairwise and does not depend on the observation modality or policy architecture. It can therefore be independently applied to every component policy whose prediction participates in the current composition. At composition time t_c, we choose a common target frame T_c, typically the action frame associated with the most recently queried policy. Let $\tau^{(m)} \le t_c$ denote the query time of the latest available prediction from policy $\pi^{(m)}$, and let $T_m = T(\tau^{(m)})$ denote its associated action frame. For each policy, we select the portion of its guidance that overlaps with the composed horizon and independently rebase it into T_c:

$$\epsilon_k'^{(m)} = \mathbf{A}_{c\leftarrow m} S(\epsilon_k^{(m)}), \qquad m \in \mathcal{M}_c, \quad (11)$$

where $\mathcal{M}_c$ is the set of component policies available for composition at t_c. After this transformation, all selected guidance terms share the same temporal support and action reference frame.

The aligned guidance terms can then be composed through a normalized weighted combination:

$$\epsilon_{k,i}^{\text{fuse}} = \sum_{m\in\mathcal{M}_{c,i}} w_i^{(m)} \epsilon_{k,i}'^{(m)} \Big/ \sum_{m\in\mathcal{M}_{c,i}} w_i^{(m)}, \quad (12)$$

where $\mathcal{M}_{c,i} \subseteq \mathcal{M}_c$ contains the policies whose predictions cover horizon step i, and $w_i^{(m)} \ge 0$ denotes the contribution of policy m at that step. The weights can reflect prediction freshness, horizon position, or component reliability.

Finally, we map the fused guidance from the common frame T_c back to the original full-horizon action frame, replace the corresponding segment of the full-horizon guidance while leaving the rest unchanged, and use the updated guidance for the DDIM update.

IV. Experiments

We answer the following research questions through experiments: **(Q1)** Does asynchronous multi-rate composition improve performance over existing composition strategies and policy architectures? **(Q2)** What roles do relative actions and reference-frame rebasing play in asynchronous composition? **(Q3)** Can MATE effectively compose heterogeneous policies across representations, viewpoints, and inference rates? **(Q4)** How does preserving policy inference rate affect composed-policy performance?

A. Setup

Tasks. We evaluate MATE on three real-world manipulation tasks. The *Flip* and *Polish Curve* tasks, shown in Fig. 4, require both global visual perception and fine-grained force feedback. The Occluded Pick-and-Place (*PnP*) task, shown in Fig. 6, requires the robot to retrieve a occluded doll from a high-walled box and place it on a designated white disk. For *Flip* and *Polish Curve*, we instantiate MATE by composing a RISE-based vision policy with a lightweight force policy, operating at inference rates of 5 Hz and 25 Hz, respectively. For *PnP*, we compose a global-camera 3D vision policy based on RISE with a wrist-camera 2D vision policy, operating at 2 Hz and 5 Hz, respectively. These component inference rates are used unless otherwise stated.

Data Collection. We perform real-world experiments on a Flexiv Rizon4 robot equipped with an FT03S six-axis force/torque sensor. For each task, we collect 50 demonstrations using an arm-to-arm teleoperation system with accurate force feedback [30]. This setup allows the operator to provide contact-rich manipulation trajectories while preserving fine-grained interaction information between the robot and the environment. During data collection, we record synchronized multimodal observations, including color images, depth images, robot proprioceptive states, and force/torque measurements. The color and depth streams are recorded at 20 Hz, while the low-dimensional data is recorded at 1000 Hz.

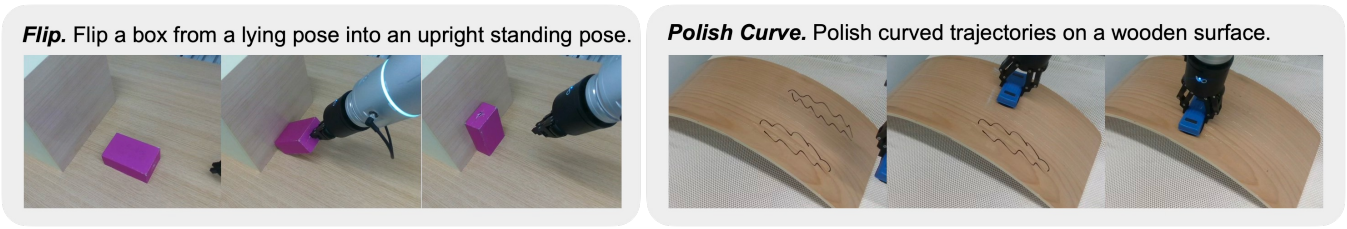


Fig. 4: Contact-Rich Manipulation Tasks. We evaluate MATE on two contact-rich tasks that require both visual perception and fine-grained force/torque feedback. Both tasks require effective coordination and adaptation across visual and force/torque modalities.

Policy	Modular	Inference Freq.	*Flip*		*Polish Curve*	
			Push	Flip	Contact	Score
RISE [37] (vision policy)	-	5 Hz	95.0%	30.0%	90.0%	35.00%
Force-only	-	25 Hz	5.0%	5.0%	25.0%	6.25%
Training Noise Fusion	✗	5 Hz	30.0%	17.5%	70.0%	20.00%
Synchronous Composition	✓	5 Hz	**100.0%**	25.0%	**100.0%**	25.00%
Policy Consensus [6]	✓†	5 Hz	95.0%	40.0%	**100.0%**	42.50%
Absolute Action	✓	25 Hz	60.0%	12.5%	75.0%	21.25%
Relative Action w/o Rebasing	✓	25 Hz	90.0%	47.5%	**100.0%**	37.50%
TA-VLA [45]	✗	5 Hz	80.0%	30.0%	**100.0%**	47.50%
RDP [41]	✗	24 Hz	**100.0%**	35.0%	**100.0%**	45.00%
MATE (*ours*)	✓	25 Hz	95.0%	**60.0%**	**100.0%**	**55.00%**

TABLE I: Evaluation Results. MATE achieves the best final-task performance, demonstrating the effectiveness of modular, rate-preserving policy composition. Gray cells indicate high-frequency inference. The † denotes that Policy Consensus [6] requires training an additional router and fine-tuning the component policies.

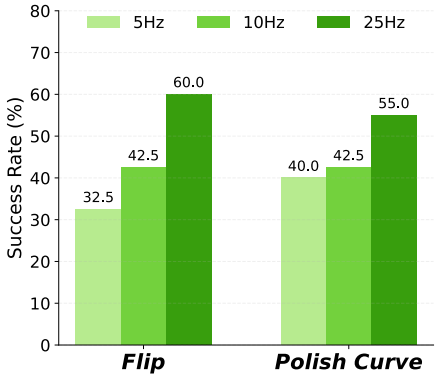


Fig. 5: Effect of Inference Rate. Final-task performance on *Flip* and *Polish Curve* at different inference rates. Higher rates enable more timely guidance and improve the composed policy on both tasks.

Baselines. To isolate the contribution of the composition design, we include five critical controls. Training-Time Noise Fusion combines diffusion noises during training, whereas Synchronous Composition combines modality-specific policy branches at synchronized inference steps. Building on Synchronous Composition, Policy Consensus [6] employs a learned router to determine the contribution of each modality. Absolute Action asynchronously composes policies that directly predict actions in a shared absolute frame. Relative Action w/o Rebasing instead represents actions relative to each policy's latest observation but directly fuses these predictions without transforming them to a common temporal reference frame. We further compare against the representative multimodal policies TA-VLA [45] and RDP [41].

Metrics. For *Flip*, "Push" measures whether the box reaches the target, and "Flip" measures successful flipping, with partial flips counted as half successes. For *Polish Curve*, "Contact" measures surface contact, and "Score" is $1.0/0.75/0.25/0$ for complete erasure, over-50% erasure, partial erasure below 50%, and failure, respectively. For *PnP*, "Reach" measures whether the robot's end effector enters the high-walled box, and "Pick" measures whether the gripper grasps the toy. "Place" measures whether the robot successfully deposits the toy on the designated white disk.

Protocols. All policies are deployed on a local workstation equipped with a NVIDIA RTX 5090 GPU, with the exception of TA-VLA [45], whose memory footprint exceeds the workstation's GPU capacity. TA-VLA [45] is therefore served on a remote node with 4 NVIDIA A100 GPUs. To ensure a fair and reproducible comparison, we adopt an identical evaluation protocol across all policies. We pre-sample a fixed set of initial object placements that are uniformly distributed over the workspace, and reuse the same set of placements for every policy. This isolates the effect of the policy itself from variation in the test layout, so that all methods are evaluated under identical initial conditions. Unless otherwise stated, the main results use the seen object and randomize only its placement. Each policy is evaluated over 20 consecutive trials per task.

B. Vision-Force Policy Composition

Asynchronous multi-rate composition improves over existing composition strategies (Q1). Table I shows that MATE consistently outperforms other fusion methods. For training-time noise fusion, high-frequency force inputs are forced into temporal alignment with slower vision observations, limiting their capability. Compared with synchronous composition, this gap suggests that bottlenecking all modalities at slow inference frequency limits the utilization of modality-specific capabilities, especially for lightweight, high-frequency modalities such as force feedback. Policy Consensus further adds a router on top of synchronous composition, but still underperforms MATE. We found that the learned router weights may overfit to one dominant modality during training, leading to imbalanced modality usage. Instead, MATE asynchronously composes modality-specific policies at inference time, allowing each modality branch to operate at its native frequency rather than being

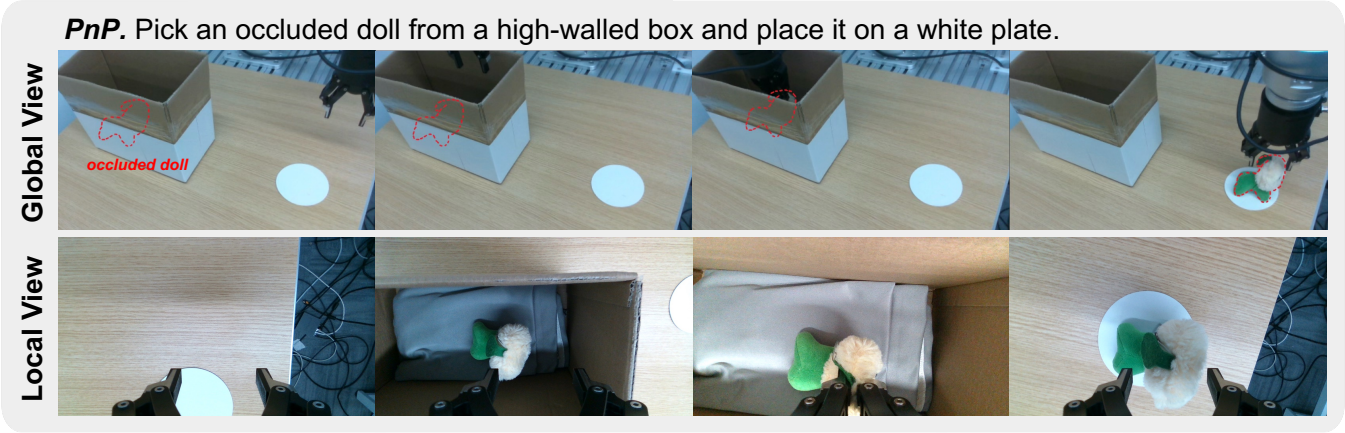


Fig. 6: Occluded Pick and Place (*PnP*) Task. The robot reaches into a high-walled box, grasps a doll, and places it on a designated white disk. The box walls frequently occlude the doll from the global camera, requiring complementary close-range observations from the wrist camera.

bottlenecked by the slowest modality.

MATE achieves stronger overall performance than existing policy architectures (Q1). Standalone and jointly trained policy architectures exhibit different limitations on these contact-rich tasks. The vision-only RISE backbone lacks contact-aware feedback and struggles during contact-sensitive stages. RDP adopts a specially designed jointly trained multimodal architecture, but does not explicitly align and compose modality-specific guidance at inference time. TA-VLA is limited by its low inference speed, which restricts reactive correction using force feedback. By contrast, MATE combines long-horizon visual guidance with high-frequency force feedback, enabling more responsive contact adaptation and stronger overall task performance.

Relative actions and reference-frame rebasing enable consistent asynchronous composition (Q2). Predicting absolute actions substantially degrades performance because the force-only policy receives no proprioceptive state: force observations describe local contact but do not determine the robot's global end-effector pose, making absolute action prediction ambiguous. We therefore predict actions relative to the end-effector frame for all policies. However, asynchronous policies produce these relative actions at different robot poses, so directly averaging them without rebasing composes geometrically inconsistent guidance. Although this operation is mathematically inconsistent, its empirical degradation is moderate at the tested 5 Hz vision policy frequency because the robot typically undergoes limited motion during the short interval, resulting in a relatively small reference-frame mismatch (Table I). Rebasing transforms the asynchronous predictions into a common current reference frame before fusion, enabling consistent multi-rate composition.

C. 3D-2D Vision Policy Composition

MATE composes heterogeneous policies across representations, viewpoints, and inference rates (Q3). We further evaluate MATE in a heterogeneous visual-policy setting, where the two component policies differ in representation, viewpoint, and inference rate. On the *PnP* task shown in Fig. 6, the global-camera 3D policy provides global geometric guidance but often loses sight of the toy because of occlusion by the box walls. Consequently, RISE can approach the box yet often fails to reliably localize and grasp the toy. The wrist-camera 2D policy provides complementary close-range observations after the end effector enters the box. MATE combines global 3D guidance with local 2D observations to support occlusion-sensitive manipulation. As shown in Table II, this composition improves execution across the Reach, Pick, and Place stages, particularly after the robot enters the box.

Policy	Infer Freq.	*PnP*		
		Reach	Pick	Place
RISE [37]	2 Hz	60.0%	15.0%	10.0%
Training Noise Fusion	2 Hz	95.0%	85.0%	80.0%
Sync. Composition	2 Hz	90.0%	80.0%	75.0%
Policy Consensus [6]	2 Hz	**100.0%**	85.0%	80.0%
MATE (*ours*)	5 Hz	**100.0%**	**90.0%**	**85.0%**

TABLE II: Evaluation Results on *PnP* task. MATE achieves the best overall performance across the three stages of the *PnP* task. The cell highlighted in gray indicates high-frequency inference.

Asynchronous composition provides modest but consistent gains over alternative fusion strategies. MATE achieved the best results across all three stages, although its improvement over the fusion baselines was relatively small. This suggests that increasing the inference frequency is less critical in this task than effectively combining complementary policy predictions. Nevertheless, MATE preserves the faster policy's inference rate without sacrificing slower global guidance, demonstrating that the framework is not tied to a particular sensing modality or policy representation.

D. Effect of Policy Inference Frequency

Preserving higher policy inference rates improves composed-policy performance (Q4). We evaluate MATE under multiple force policy inference rates on the *Flip* and *Polish Curve* tasks. Across these settings, the vision policy maintains a fixed, lower inference rate, while the force policy inference rate is varied. As shown in Fig. 5, increasing the force policy inference rate improves final task success on

both tasks. This trend suggests that more frequent incorporation of updated contact feedback contributes to the performance gains while complementing the lower-rate global visual guidance. In both tasks, longer intervals between policy updates may allow contact errors to accumulate before corrective actions are generated. A faster force policy can shorten this feedback-to-correction interval, while the slower vision policy continues to provide global task guidance. These results indicate that synchronizing both policies at the slower inference rate limits the contribution of the force policy. Within the evaluated range, preserving heterogeneous inference rates improves the use of complementary policies, supporting the rate-preserving design of MATE.

V. Conclusion

We presented MATE, a modular framework for asynchronous multi-rate diffusion policy composition. MATE allows independently trained component policies with heterogeneous observation representations and inference rates to contribute guidance without being synchronized to the slowest component. To enable consistent asynchronous composition, we introduced temporal overlap selection and derived a reference-frame rebasing rule that aligns diffusion variables generated at different robot states before weighted fusion. We instantiated MATE by composing a lower-rate vision policy with a higher-rate force policy on two contact-rich manipulation tasks, and further demonstrated its generality through the composition of complementary 3D and 2D visual policies on an occluded pick-and-place task. Across these real-world tasks, MATE outperforms synchronous composition, existing vision-only and multimodal policies, and a matched-frequency end-to-end baseline. Our frequency analysis confirms that preserving native inference rates benefits composed policies. Overall, MATE provides a modular framework for composing diffusion policies at heterogeneous inference rates across different modalities, representations, and viewpoints.

Limitations and Future Work. Although MATE improves over existing baselines, there is still room to further enhance task success and control stability. Our current framework is developed for diffusion-based policies, extending it to flow matching [2, 3] or other generative action modeling frameworks remains future work. Moreover, our experiments focus on parallel-gripper manipulation. Future work could apply MATE to dexterous hands [18, 34], where richer contact dynamics, higher-dimensional actions, and the need for high-frequency reactive control may further highlight the benefits of asynchronous multi-rate policy composition.

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APPENDIX

VI. Method Implementation Details

A. Visual Policy Details

Our 3D visual policy follows the original RISE design [37]. We retain its point-cloud input representation, network architecture, diffusion action prediction head, and training procedure. The main modification is the action coordinate representation: instead of predicting actions in the camera coordinate frame, we represent actions relative to the end-effector pose at the policy query time. This representation allows predictions generated at different query times to be rebased into a common action reference frame before asynchronous composition. All other hyperparameters and implementation choices of the 3D visual policy follow RISE.

For the *Occluded Pick-and-Place* task, we additionally train a standard RGB-conditioned Diffusion Policy using observations from the wrist camera. The policy takes a single RGB image at a resolution of 720×1280 and encodes it into a feature using ResNet-18. Both vision policies predict a 100-step action chunk at 50 Hz, corresponding to a 2 s prediction horizon. Each action uses a relative representation with 10 dimensions: 3D translation, a column-wise 6D rotation representation, and a gripper command.

B. Force Policy Details

We train a force-conditioned diffusion policy as the low-dimensional force policy. In the main compositional policy, the force policy takes a 1s force/torque history as input, corresponding to 100 steps at 100Hz with six force/torque channels, and predicts a 1s future action chunk, corresponding to 50 action steps at 50Hz. Each action is represented in a 10D space, including 3D translation, 6D rotation, and one gripper command. Rotations are parameterized by the column-wise 6D rotation representation. The force/torque stream is expressed in the TCP frame and normalized to $[-1, 1]$ using fixed per-axis force/torque bounds. Actions are represented in a relative formulation, using the current end-effector pose as the base pose. The key hyperparameters of the force policy are summarized in Table III.

For the standalone force policy analysis in Appendix VII-B, we additionally vary the force/torque input frequency and inference frequency to study their effects on performance.

Item	Setting
Force input length	1s
Force input frequency	100Hz
Force input steps	100
Force/torque channels	6
Force feature dimension	16
Force encoder	MLP: $600 \to 256 \to 16$
Action chunk length	50
Action chunk frequency	50Hz
Action representation	10D: 3D translation, 6D rotation, gripper
U-Net channels	$[32, 64]$
DDPM training steps	100
DDIM inference steps	10
Optimizer	AdamW
Learning rate	3×10^{-4}
Batch size	256
Epochs	100

TABLE III: Key Hyperparameters of the Force Policy. The table reports the default configuration with a 100 Hz force/torque input frequency. In the standalone frequency study, the input dimension of the force encoder is adjusted according to the input frequency, while the remaining architecture and training settings are kept consistent unless otherwise specified.

C. Inference and Asynchronous Composition Details

Horizon-Dependent Composition Weights. During asynchronous composition, the vision and force/torque policies provide guidance at different update rates. The vision policy is queried less frequently and provides global, long-horizon guidance, while the force/torque policy is queried more frequently and provides fresher local guidance for contact-sensitive correction. Therefore, when fusing the two guidance signals, we use a latency-aware weighting schedule that changes along the selected action horizon.

Specifically, for each action index within the force policy horizon, we compute the vision weight by linearly decreasing it from a high value to a low value. Let i denote the action index and $I_{\max}$ denote the maximum index of the selected horizon. We first compute a normalized horizon ratio $r_i = i/I_{\max}$. The vision weight is then computed by linear interpolation:

$$w_i^V = w_{\text{start}}^V + r_i \left(w_{\text{end}}^V - w_{\text{start}}^V\right). \tag{13}$$

In our implementation for vision–force composition, $w_{\text{start}}^V = 0.8$, $w_{\text{end}}^V = 0.2$, and $I_{\max} = 50$. Therefore, the vision weight starts from 0.8 at the beginning of the selected horizon and gradually decreases to 0.2 by the end of the horizon. The force weight is defined as the complementary value $w_i^F = 1 - w_i^V$. This creates an index-dependent weighting schedule: early actions rely more on vision guidance, while later actions rely more on force/torque guidance.

Runtime Analysis. We report the runtime of the proposed asynchronous vision–force composition during evaluation. With our implementation, the mean policy inference time after composition is approximately 40ms, which enables 25Hz policy inference. In contrast to synchronous composition, where all modalities are typically constrained by the slowest policy, our asynchronous formulation allows the low-dimensional force branch to be updated at a higher frequency while still incorporating visual guidance when available.

VII. Experiment Details

A. Setup

Data Collection. We visualize the initial configurations of all 50 demonstrations for each task in Fig. 8. The objects are randomly placed within the workspace during demonstration collection.

Evaluation Protocols. To ensure a fair and reproducible comparison, we adopt an identical evaluation protocol across all policies. As shown in Fig. 9, we pre-sample a fixed set of

Object					
Size(cm)	14.2*9.0*5.1	9.0*14.2*5.1	11.4*11.4*6.0	10.0*6.9*6.9	6.0*6.0*21.0
Weight(g)	128.6	90.8	110.9	239.4	270.0

Fig. 7: Evaluation Objects for the Low-dimensional Force Policy on *Flip*. The policy is tested on five objects with varying sizes and weights. Only the first purple box is seen during training, while the other four objects are unseen.

initial object placements that are uniformly distributed over the workspace, and reuse exactly the same set of placements for every policy. This isolates the effect of the policy itself from variation in the test layout, so that all methods are evaluated under identical initial conditions. Unless otherwise stated, the main results in Table I use the seen object and randomize only its placement. Each policy is evaluated over 20 consecutive trials per task, and success rates are computed accordingly.

B. Additional Experiments Results

We additionally evaluate the standalone low-dimensional force/torque policy to analyze the effect of input and inference frequencies, as motivated in the teaser and Introduction. Unlike the full-task evaluation in Table I, this experiment initializes the robot near the object in a pre-contact configuration, isolating the contact-sensitive flipping stage to study the impact of inference frequency. Fig. 10 shows the evaluation setup. The policy is tested on five objects, including the *Purple Box* used in training and four unseen objects with different shapes, materials, and contact properties. This controlled setup allows us to systematically evaluate how input and inference frequencies affect contact-sensitive manipulation.

We further report the detailed success rates of the standalone force policy under different input-frequency and inference-frequency configurations in Fig. 11. For each input frequency, we keep the force history window fixed to 1s and vary the number of input steps according to the input frequency. We evaluate each policy with three inference frequencies, 2Hz, 5Hz, and 25Hz, across 5 objects. Each input-frequency and inference-frequency configuration is evaluated with 10 trials per object. The results show that increasing the inference frequency generally improves the success rate, especially when the input frequency is low. These results support our motivation that force feedback should not be bottlenecked by a low inference rate, and that high-frequency local feedback is important for contact-sensitive manipulation.

C. Failure Analysis

We further analyze representative failure modes for the *Flip* and *Polish Curve* tasks. Overall, the failures are closely related to insufficient reactivity to force feedback and the difficulty of balancing global visual guidance with local contact feedback.

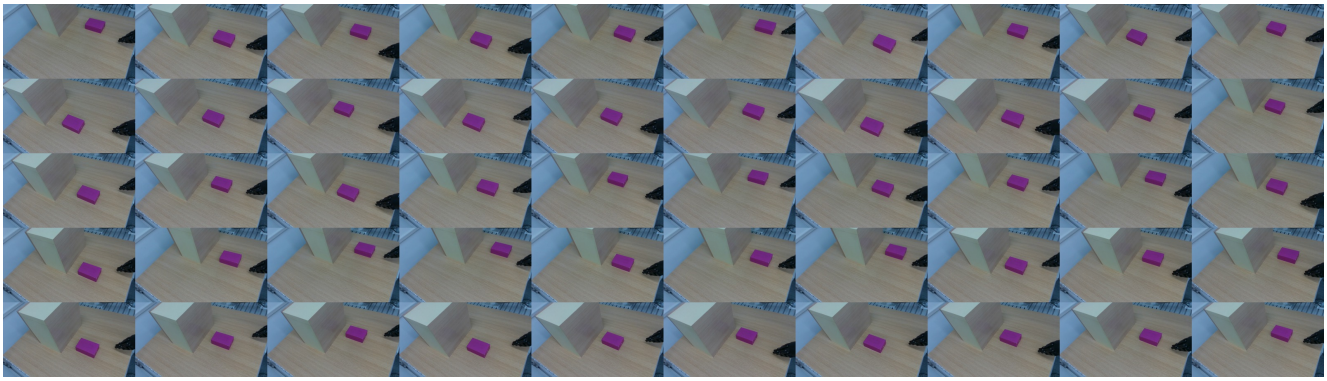
Flip. For *Flip*, we observe three common failure modes in the baselines. First, for force based baselines, the force signal can be noisy during the contact-free pushing stage. This noise may cause the robot arm to drift from the desired contact position with the small box, making the subsequent flipping motion difficult to complete. Second, during the flipping stage, some baselines do not run at a sufficiently high inference frequency and therefore cannot react quickly to force feedback. As a result, the robot fails to adjust its end-effector position in time after contact, leading to missed or incomplete flips. Third, vision-only policies do not have access to force feedback and may prematurely switch to the flipping motion before the small box establishes contact with the target box.

Polish Curve. For *Polish Curve*, the main challenge is to maintain stable and sufficiently strong contact while following the curved trajectory. Baselines with low inference frequency often fail to consistently maintain large contact force, resulting in weak or unstable polishing behavior. Vision-only policies may also enter the polishing stage before actual contact is established, since they cannot directly sense the contact state.

Compared with these baselines, MATE better balances visual and force guidance. The visual policy provides global guidance, while the force/torque policy provides local guidance for contact-sensitive correction. And it allows each modalities run at its native frequency. This makes the composed policy more reactive during contact transitions and more stable during sustained contact, leading to stronger performance on both tasks.

Occluded Pick-and-Place (PnP). For *PnP*, the main challenge arises from severe object occlusion caused by the high-walled box. The global-camera 3D policy provides useful guidance for reaching the box but often fails to localize the toy once the robot enters the box, resulting in unsuccessful grasps and subsequent placement failures. Although the wrist-camera 2D policy provides complementary close-range observations, effectively coordinating global spatial guidance with local visual feedback remains challenging. As a result, existing fusion baselines still exhibit failures during grasping and placement. MATE mitigates these limitations by asynchronously composing the two visual policies, preserving global guidance while incorporating more frequent local observations. Nevertheless, occasional failures remain during the picking and placing stages, indicating room for further improvement in occlusion-sensitive manipulation.

Flip



Polish Curve

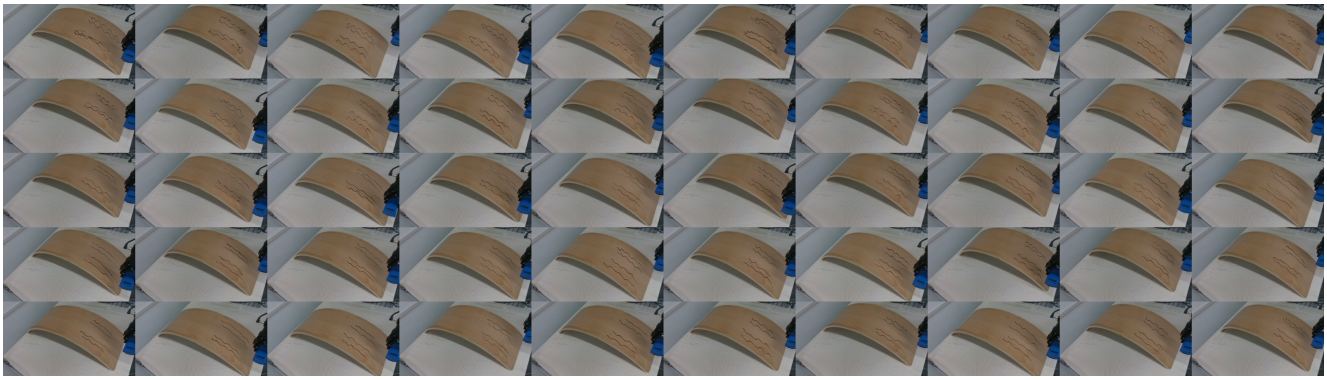
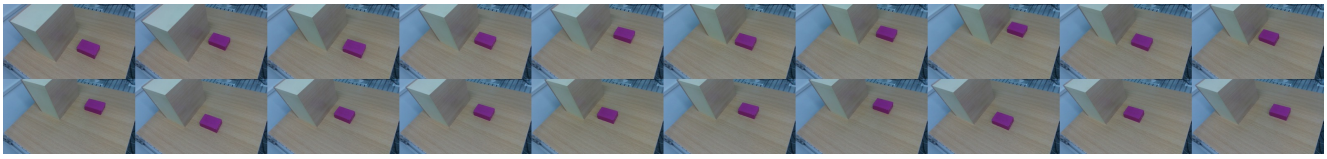


Fig. 8: Overview of Demonstrations for each Tasks. We visualize the initial configurations from 50 demonstrations for each task.

Flip



Polish Curve



Fig. 9: Evaluation Initial Configurations. For each task, we evaluate each policy on 20 different initial configurations. The object is randomly placed at the beginning of each trial, and these evaluation configurations are different from those used in the training demonstrations.

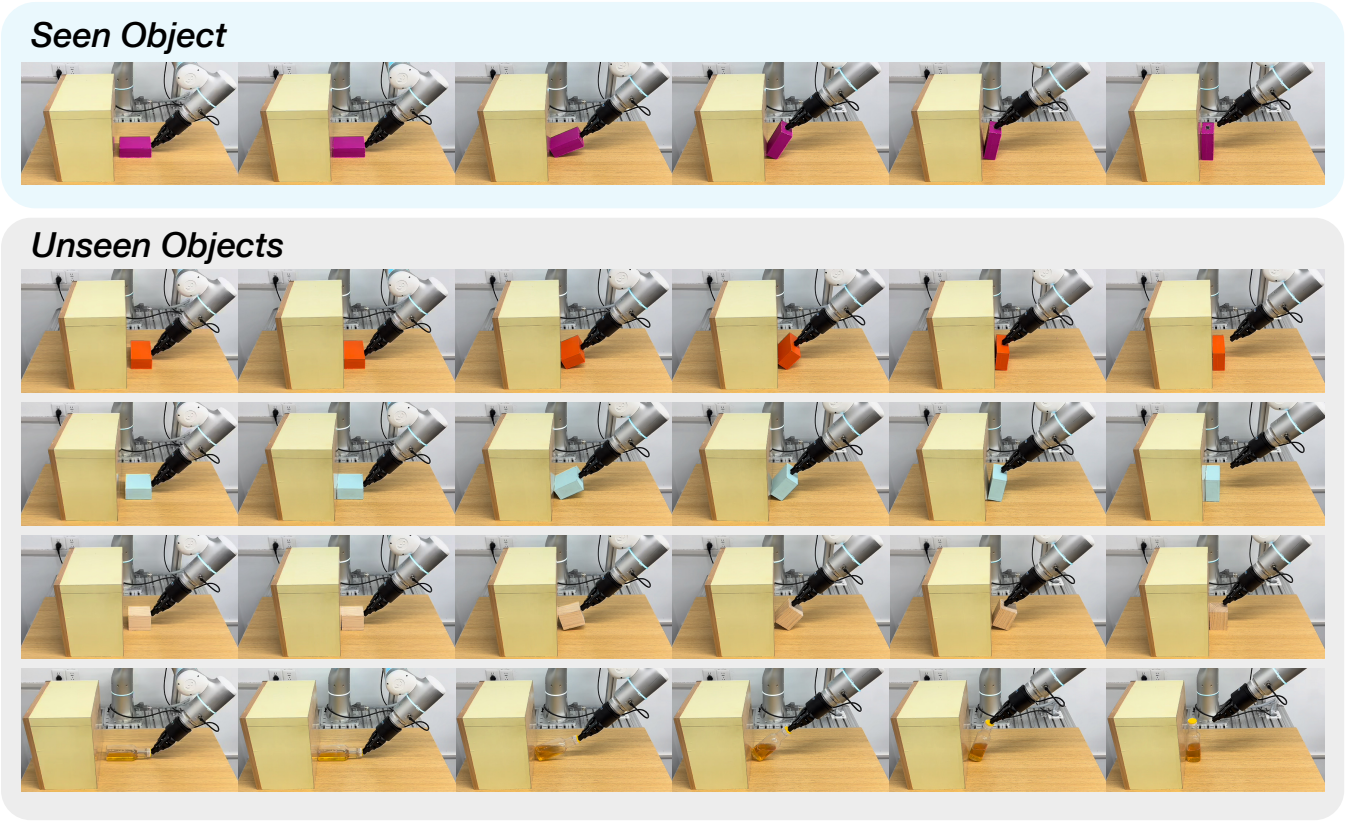


Fig. 10: Evaluation Setup for the Standalone Low-dimensional Policy. We evaluate the standalone low-dimensional force/torque policy on five objects with different shapes, materials, and contact properties. The top row shows the *Purple Box* used in training, while the bottom rows show additional test objects used during evaluation. This setup is used to compare the standalone policy under different input-frequency and inference-frequency configurations.

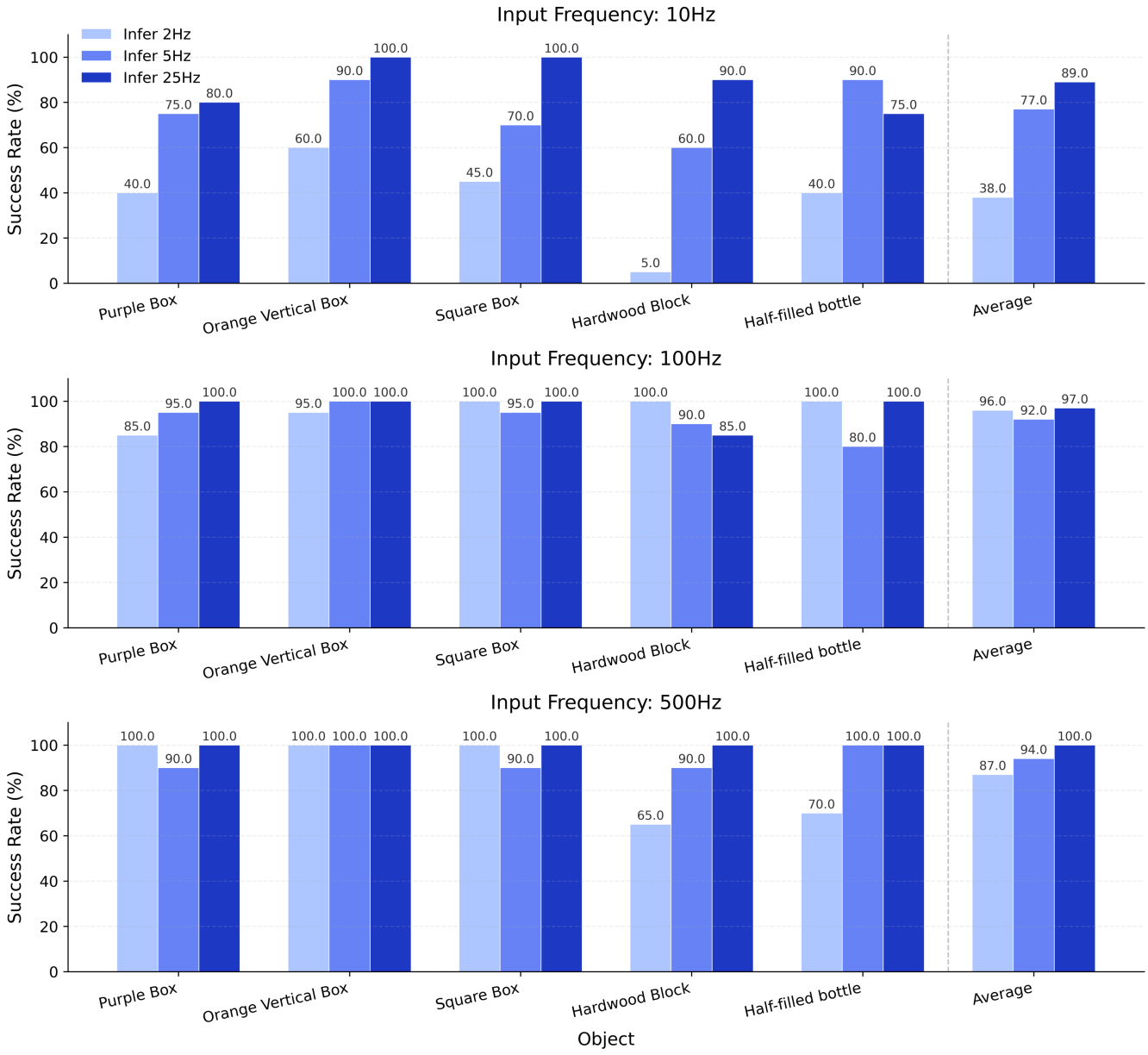


Fig. 11: Detailed Standalone Force Policy Results under Different Input and Inference Frequencies. We evaluate force/torque policies trained with different input frequencies, including 10Hz, 100Hz, and 500Hz, and test each policy under different inference frequencies, including 2Hz, 5Hz, and 25Hz. Results are reported across five test objects, with the average success rate shown on the right of each subplot. Increasing the inference frequency generally improves success rate, especially under low-frequency input, while high-frequency input combined with high-frequency inference achieves the best overall performance.